SYNTHETIC · SIMPLIFIED MODEL

TurbineIQ

A clear view of model versus synthetic operation.

No physical connection.

A USEFUL COMPARISON

Where does synthetic data diverge from a fixed reference?

And which component helps explain the gap?

REFERENCE → OBSERVATION → COMPONENT

Synthetic efficiency parameter
82.5 %
Known synthetic scenario truth

Discharge temperature
649.4 K
Synthetic plant-model state

Compressor power
36143.51 kW
Synthetic plant-model state

Metric ledger

Sample 49 · 12:00 UTC

Metric	Synthetic	Reference	Delta	Provenance
Synthetic efficiency parameter	82.5 %	85.0 %	-2.5 %	Known synthetic scenario truth
Discharge temperature	649.4 K	638.7 K	+10.7 K	Synthetic plant-model state
Compressor power	36143.51 kW	35069.27 kW	+1074.23 kW	Synthetic plant-model state

TWO VALUES · ONE SIGNED RESIDUAL

Reference vs observation

The model holds its assumptions fixed; the seeded synthetic sample supplies the observation.



Reference model

Synthetic observation

Observation – reference

Values remain tied to the selected sample

Seed 42

COMPRESSOR-PERFORMANCE DRIFT

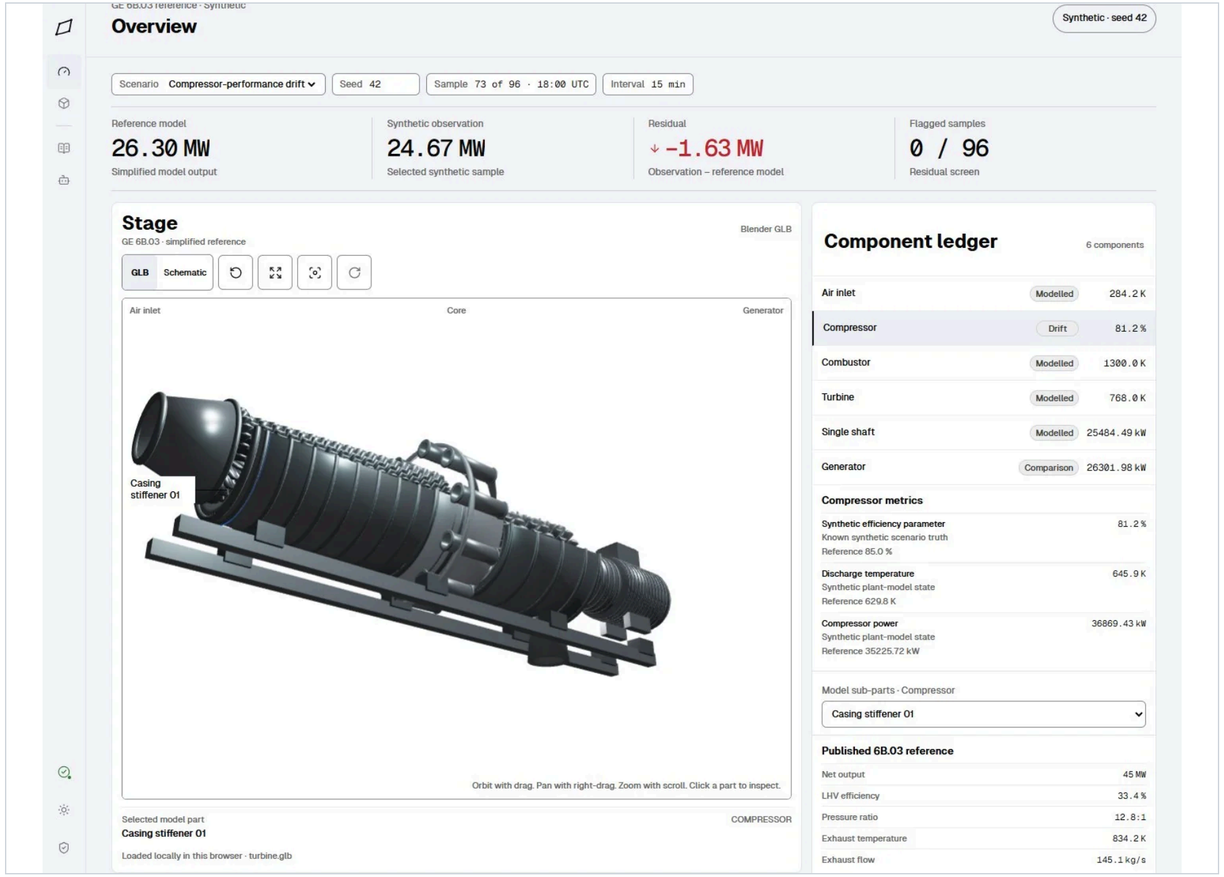
One sample, one comparison.

26.30 MW
Reference model

24.67 MW
Synthetic observation

−1.63 MW
Residual · observation minus reference

Seed 42 · sample 73 of 96 · 0 of 96 flagged



INTERACTIVE STAGE

Inspect one component in 3D.

The selected compressor mesh stays in the context of the whole model.

Selected mesh · Casing stiffener 01



LOCAL LEXICAL RETRIEVAL

Every passage keeps its citation.

Search and inspect a source excerpt without sending it to a model.

Reader `[demo_cycle_notes.md, section "Simplified cycle"]`

The model follows four states in an open Brayton cycle. State 1 is ambient air at the compressor inlet. State 2 is compressor discharge. State 3 is the turbine inlet after approximate heat addition and combustor pressure loss. State 4 is exhaust at ambient pressure.

For a constant specific heat ratio gamma, the ideal compressor temperature is T2s = T1 * pressure_ratio^((gamma - 1)/gamma). The actual compressor temperature is T2 = T1 + (T2s - T1)/compressor_efficiency. Compressor specific work is cp_air * (T2 - T1).

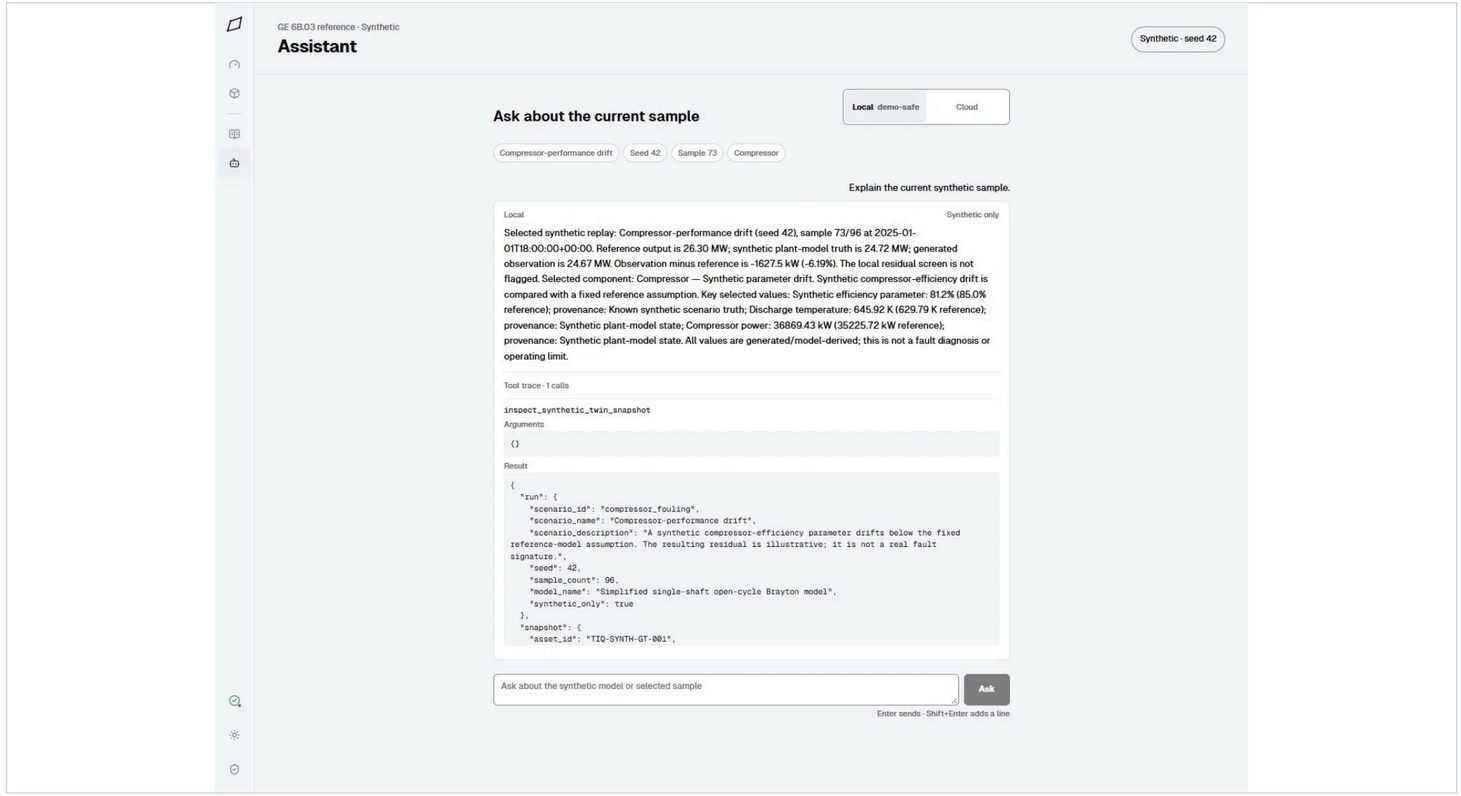
The ideal turbine exhaust temperature is T4s = T3 * (P4/P3)^((gamma_gas - 1)/gamma_gas). Actual exhaust temperature is T4 = T3 - turbine_efficiency * (T3 - T4s). Turbine specific work is approximated as (1 + fuel_to_air_ratio) * cp_gas * (T3 - T4).

Source	demo_cycle_notes.md
Section	Simplified cycle
Retrieval	local lexical retrieval

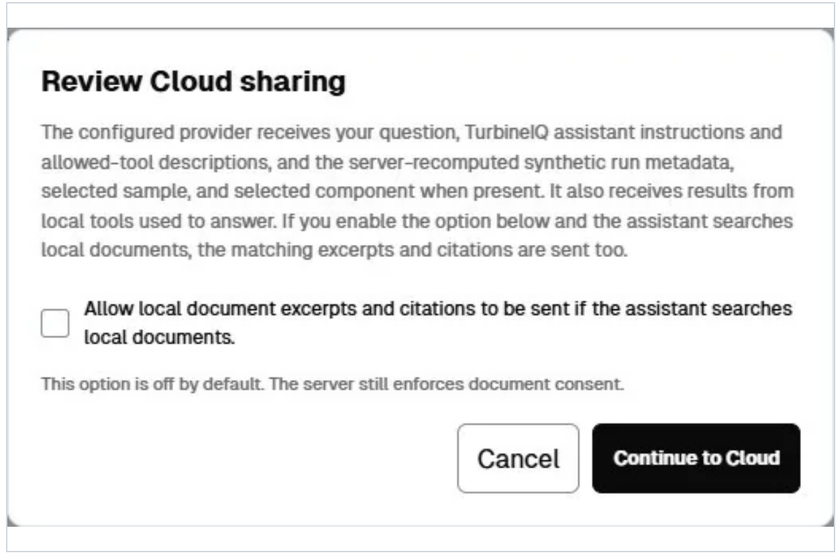
LOCAL FIRST · CLOUD BY CONSENT

Context is visible. Tool calls are inspectable.

Cloud mode opens a clear sharing sheet first; document excerpts remain a separate opt-in.



Local answer · expanded tool trace



Cloud consent · excerpts off by default

Server-recomputed context

No cloud call in this demo

SYNTHETIC · SIMPLIFIED MODEL

Keep the boundary clear.

Synthetic data only · no real-time connection or control.

Next: review assumptions and component names with turbine engineers.